Lösung: Ableitungen mit dem Differentialquotient

a) $f(x) = ax+b$

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{a\cdot(x+\Delta x)+b-(ax+b)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{ax+a\Delta x+b-ax-b}{\Delta x} = \lim_{\Delta x \to 0} \frac{a\Delta x}{\Delta x}$$

$$= \lim_{\Delta x \to 0} a = \underline{\underline{a}}$$

b) $f(x) = \sqrt{x}$

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{\sqrt{x+\Delta x} - \sqrt{x}}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{\sqrt{x+\Delta x} - \sqrt{x}}{\Delta x} \cdot \frac{\sqrt{x+\Delta x} + \sqrt{x}}{\sqrt{x+\Delta x} + \sqrt{x}}$$

3. Binomische Formel

$$= \lim_{\Delta x \to 0} \frac{\sqrt{x+\Delta x}^2 - \sqrt{x}^2}{\Delta x\,(\sqrt{x+\Delta x} + \sqrt{x})}$$

$$= \lim_{\Delta x \to 0} \frac{x+\Delta x - x}{\Delta x\,(\sqrt{x+\Delta x} + \sqrt{x})} = \lim_{\Delta x \to 0} \frac{\Delta x}{\Delta x\,(\sqrt{x+\Delta x} + \sqrt{x})}$$

$$= \lim_{\Delta x \to 0} \frac{1}{\sqrt{x+\Delta x} + \sqrt{x}} = \frac{1}{\sqrt{x} + \sqrt{x}} = \underline{\underline{\frac{1}{2\sqrt{x}}}}$$